Git and LaTeX Guide

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1 Git and GitHub

1.1 Downloading and Installing Git Bash on Windows

1.1.1 What is Git Bash?

Git Bash is a terminal application for Windows that lets you run Git commands (and many Unix-style commands like `ls`, `cd`, `ssh-keygen`) using a Bash shell, instead of the default Windows Command Prompt.

1.1.2 Download the Installer

1. Open a web browser and go to `https://git-scm.com/downloads`.
2. Click **Windows** under the list of operating systems.
3. The download should start automatically. If not, click the manual download link for the latest 64-bit version.

1.1.3 Run the Installer

1. Locate the downloaded file (usually named something like `Git-2.xx.x-64-bit.exe`) in your **Downloads** folder and double-click it.
2. If prompted by Windows "Do you want to allow this app to make changes to your device?", click **Yes**.
3. Click **Next** on the license screen.

1.1.4 Installation Options

The installer presents several configuration screens. For most beginners, the default options are fine — just keep clicking **Next**. A few worth knowing about:

- **Select Components**: default checkboxes are fine.
- **Choosing the default editor**: pick whichever text editor you're comfortable with (Notepad is a safe choice for beginners).
- **Adjusting your PATH environment**: choose "Git from the command line and also from 3rd-party software" (this is usually the recommended/default option).
- **Choosing the SSH executable**: use the bundled OpenSSH (default option).
- **Configuring the line ending conversions**: keep the default ("Checkout Windows-style, commit Unix-style line endings").
- **Configuring terminal emulator**: use MinTTY (default option).

1.1.5 Finish Installation

1. Click **Install** and wait for the process to complete.
2. Click **Finish**.

1.1.6 Open Git Bash

1. Right-click anywhere on your Desktop or inside a folder, and select **Git Bash Here** from the context menu.
2. Alternatively, click the **Start** menu, search for "Git Bash", and open the application.

1.1.7 Verify the Installation

In the Git Bash window that opens, type:

```
git --version
```

You should see output similar to:

```
git version 2.45.1.windows.1
```

If a version number appears, Git Bash is installed correctly and you're ready to use commands like `git clone`, `ssh-keygen`, and `ssh -T git@github.com`.

1.2 Adding an SSH Key to GitHub and Cloning a Repository

1.2.1 Check for Existing SSH Keys

Open a git bash terminal and check whether you already have an SSH key pair.

```
ls -al ~/.ssh
```

Look for files named `id_ed25519` / `id_ed25519.pub` or `id_rsa` / `id_rsa.pub`. If these exist, skip to the "Add the Key to GitHub" step.

1.2.2 Generate a New SSH Key

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

1.2.3 Start the SSH Agent and Add Your Key

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
```

1.2.4 Copy the Public Key

```
cat ~/.ssh/id_ed25519.pub
```

1.2.5 Add the Key to GitHub

1. Log in to GitHub and go to **Settings**.
2. Click **SSH and GPG keys** in the sidebar.
3. Click **New SSH key**, give it a title, and paste the key.
4. Click **Add SSH key**.

1.2.6 Test the Connection

```
ssh -T git@github.com
```

1.2.7 Clone a Repository Using SSH

```
git clone git@github.com:username/repository.git
```

1.3 Creating a New GitHub Repository from an Existing Folder

1.3.1 Overview

This guide covers turning a folder you already have on your computer (with existing files) into a Git repository, then pushing it to a brand-new repository on GitHub.

1.3.2 Step 1: Open Git Bash in the Folder

1. Navigate to the folder in Windows File Explorer.
2. Right-click inside the folder and select **Git Bash Here**.

Alternatively, open Git Bash and use `cd` to navigate there:

```
cd /c/Users/yourname/Documents/your-project
```

1.3.3 Step 2: Initialize the Folder as a Git Repository

```
git init
```

This creates a hidden `.git` folder inside your project, which is what turns an ordinary folder into a Git repository.

1.3.4 Step 3: Check the Status

```
git status
```

This lists all the files in the folder as "untracked" — Git sees them but isn't tracking changes to them yet.

1.3.5 Step 4: Stage and Commit Your Files

```
git add .
```

The `.` stages every file in the folder (and subfolders) for the next commit.

```
git commit -m "Initial commit"
```

The `-m` flag lets you attach a short message describing the commit, in quotes.

1.3.6 Step 5: Set Your Default Branch Name (Optional)

Modern GitHub repositories default to a branch named `main`. If your local repo defaulted to `master` instead, rename it:

```
git branch -M main
```

1.3.7 Step 6: Create a New, Empty Repository on GitHub

1. Log in to GitHub and click the + icon in the top-right corner, then select **New repository**.
2. Enter a **Repository name**.
3. Choose **Public** or **Private**.
4. **Important:** Do *not* check any of the boxes to add a README, `.gitignore`, or license. Since you already have files locally, starting the GitHub repo empty avoids merge conflicts.
5. Click **Create repository**.

1.3.8 Step 7: Link Your Local Folder to the GitHub Repository

After creating the repository, GitHub displays a remote URL. Copy the **SSH** URL (it should look like `git@github.com:username/repository.git`) and run:

```
git remote add origin git@github.com:username/repository.git
```

Replace `username/repository` with your actual GitHub username and repository name.

1.3.9 Step 8: Push Your Files to GitHub

```
git push -u origin main (or whatever you called it)
```

The `-u` flag sets `origin main` as the default remote and branch, so future pushes only require:

```
git push
```

1.3.10 Step 9: Verify

Refresh the repository page on GitHub. Your files should now appear there, matching what's in your local folder.

1.3.11 Note on SSH

This guide assumes SSH is already set up between your computer and GitHub. If `git push` asks for a username and password instead of working automatically, you'll need to add an SSH key to your GitHub account first.

2 Downloading and Installing Texmaker

2.1 What is Texmaker?

Texmaker is a free, cross-platform editor for writing and compiling LaTeX documents. It includes a built-in PDF viewer, syntax highlighting, and one-click compilation, making it a popular choice for beginners.

2.2 Download the Installer

1. Open a web browser and go to `https://www.xm1math.net/texmaker/download.html`.
2. Choose the installer matching your operating system (Windows, macOS, or Linux).
3. Click the corresponding download link and save the file.

2.3 Run the Texmaker Installer

1. Locate the downloaded installer file and double-click it.
2. If prompted by your operating system for permission to make changes, allow it.
3. Follow the on-screen prompts, keeping the default options, and click **Install** (or **Next**) until the installation completes.
4. Click **Finish**.

2.4 Open Texmaker

1. **Windows**: click the **Start** menu, search for “Texmaker”, and open the application.
2. **macOS**: open **Launchpad** or **Applications** and click **Texmaker**.
3. **Linux**: search for “Texmaker” in your application menu, or launch it from a terminal:

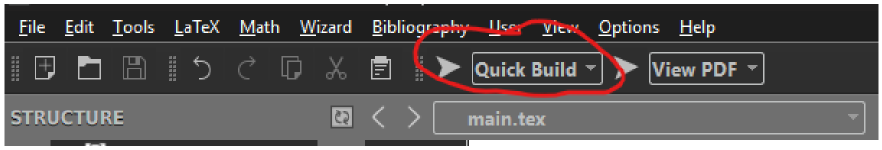
```
texmaker
```

2.5 Verify the Installation

1. In Texmaker, create a new document (**File** → **New**).
2. Type a simple test document:

```
\documentclass{article}
\begin{document}
Hello, world!
\end{document}
```

1. Save the file with a `.tex` extension.
2. Click the **QuickBuild** button (or press F1) to compile it.



If a PDF preview opens showing "Hello, world!", Texmaker and your LaTeX distribution are installed correctly.